

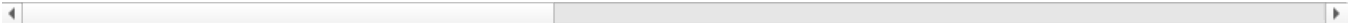
```
In [3]: import pandas as pd

In [7]: df = pd.read_csv('Telco-Customer-Churn.csv')
df

Out[7]:
```

|      | customerID | gender | SeniorCitizen | Partner | Dependents | tenure | PhoneService | MultipleLines    | InternetService | OnlineSecurity |
|------|------------|--------|---------------|---------|------------|--------|--------------|------------------|-----------------|----------------|
| 0    | 7590-VHVEG | Female | 0             | Yes     | No         | 1      | No           | No phone service | DSL             | No             |
| 1    | 5575-GNVDE | Male   | 0             | No      | No         | 34     | Yes          | No               | DSL             | Yes            |
| 2    | 3668-QPYBK | Male   | 0             | No      | No         | 2      | Yes          | No               | DSL             | Yes            |
| 3    | 7795-CFOCW | Male   | 0             | No      | No         | 45     | No           | No phone service | DSL             | Yes            |
| 4    | 9237-HQITU | Female | 0             | No      | No         | 2      | Yes          | No               | Fiber optic     | No             |
| ...  | ...        | ...    | ...           | ...     | ...        | ...    | ...          | ...              | ...             | ...            |
| 7038 | 6840-RESVB | Male   | 0             | Yes     | Yes        | 24     | Yes          | Yes              | DSL             | Yes            |
| 7039 | 2234-XADUH | Female | 0             | Yes     | Yes        | 72     | Yes          | Yes              | Fiber optic     | No             |
| 7040 | 4801-JZAZL | Female | 0             | Yes     | Yes        | 11     | No           | No phone service | DSL             | Yes            |
| 7041 | 8361-LTMKD | Male   | 1             | Yes     | No         | 4      | Yes          | Yes              | Fiber optic     | No             |
| 7042 | 3186-AJIEK | Male   | 0             | No      | No         | 66     | Yes          | No               | Fiber optic     | Yes            |

7043 rows × 21 columns



## 1. Load the dataset and display the first 5 rows.

```
In [10]: df.head(5)

Out[10]:
```

|   | customerID | gender | SeniorCitizen | Partner | Dependents | tenure | PhoneService | MultipleLines    | InternetService | OnlineSecurity | .. |
|---|------------|--------|---------------|---------|------------|--------|--------------|------------------|-----------------|----------------|----|
| 0 | 7590-VHVEG | Female | 0             | Yes     | No         | 1      | No           | No phone service | DSL             | No             | .. |
| 1 | 5575-GNVDE | Male   | 0             | No      | No         | 34     | Yes          | No               | DSL             | Yes            | .. |
| 2 | 3668-QPYBK | Male   | 0             | No      | No         | 2      | Yes          | No               | DSL             | Yes            | .. |
| 3 | 7795-CFOCW | Male   | 0             | No      | No         | 45     | No           | No phone service | DSL             | Yes            | .. |
| 4 | 9237-HQITU | Female | 0             | No      | No         | 2      | Yes          | No               | Fiber optic     | No             | .. |

5 rows × 21 columns



## 2. How many rows and columns does the dataset have?

```
In [20]: df.shape

Out[20]: (7043, 21)
```

## 3. What are the data types of each column?

```
In [23]: df.dtypes
```

```
Out[23]: customerID      object
gender      object
SeniorCitizen  int64
Partner      object
Dependents    object
tenure       int64
PhoneService  object
MultipleLines object
InternetService object
OnlineSecurity object
OnlineBackup  object
DeviceProtection object
TechSupport   object
StreamingTV    object
StreamingMovies object
Contract       object
PaperlessBilling object
PaymentMethod  object
MonthlyCharges float64
TotalCharges   object
Churn          object
dtype: object
```

## 4. Check for missing values in the dataset.

```
In [26]: df.isnull().sum()
```

```
Out[26]: customerID      0
gender      0
SeniorCitizen  0
Partner      0
Dependents    0
tenure       0
PhoneService  0
MultipleLines  0
InternetService  0
OnlineSecurity  0
OnlineBackup  0
DeviceProtection  0
TechSupport   0
StreamingTV    0
StreamingMovies  0
Contract       0
PaperlessBilling  0
PaymentMethod  0
MonthlyCharges  0
TotalCharges   0
Churn          0
dtype: int64
```

## 5. How many customers have churned?

```
In [29]: df.columns
```

```
Out[29]: Index(['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents',
               'tenure', 'PhoneService', 'MultipleLines', 'InternetService',
               'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport',
               'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling',
               'PaymentMethod', 'MonthlyCharges', 'TotalCharges', 'Churn'],
              dtype='object')
```

```
In [31]: df['Churn'].value_counts()
```

```
Out[31]: Churn
No      5174
Yes     1869
Name: count, dtype: int64
```

## 6. What is the percentage of male vs. female customers?

```
In [48]: df['gender'].value_counts(normalize=True)*100
```

```
Out[48]: gender
Male      50.47565
Female    49.52435
Name: proportion, dtype: float64
```

## 7. Find the average monthly charges for churned customers.

```
In [57]: df[df['Churn'] == 'Yes']['MonthlyCharges'].mean()
```

```
Out[57]: 74.44133226324237
```

## 8. How many senior citizens are there in the dataset?

```
In [60]: df.columns
```

```
Out[60]: Index(['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents',  
              'tenure', 'PhoneService', 'MultipleLines', 'InternetService',  
              'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport',  
              'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling',  
              'PaymentMethod', 'MonthlyCharges', 'TotalCharges', 'Churn'],  
              dtype='object')
```

```
In [62]: df['SeniorCitizen'].sum()
```

```
Out[62]: 1142
```

## 9. Show all customers with tenure > 50 months and MonthlyCharges > 80.

```
In [71]: df.columns
```

```
Out[71]: Index(['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents',  
              'tenure', 'PhoneService', 'MultipleLines', 'InternetService',  
              'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport',  
              'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling',  
              'PaymentMethod', 'MonthlyCharges', 'TotalCharges', 'Churn'],  
              dtype='object')
```

```
In [69]: df[(df['tenure'] > 50) & (df['MonthlyCharges'] > 80)]
```

```
Out[69]:
```

|  | customerID | gender     | SeniorCitizen | Partner | Dependents | tenure | PhoneService | MultipleLines | InternetService | OnlineSecurity |     |
|--|------------|------------|---------------|---------|------------|--------|--------------|---------------|-----------------|----------------|-----|
|  | 12         | 8091-TTVAX | Male          | 0       | Yes        | No     | 58           | Yes           | Yes             | Fiber optic    | No  |
|  | 15         | 3655-SNQYZ | Female        | 0       | Yes        | Yes    | 69           | Yes           | Yes             | Fiber optic    | Yes |
|  | 17         | 9959-WOFKT | Male          | 0       | No         | Yes    | 71           | Yes           | Yes             | Fiber optic    | Yes |
|  | 28         | 5248-YGIJN | Male          | 0       | Yes        | No     | 72           | Yes           | Yes             | DSL            | Yes |
|  | 30         | 3841-NFECX | Female        | 1       | Yes        | No     | 71           | Yes           | Yes             | Fiber optic    | Yes |
|  | ...        | ...        | ...           | ...     | ...        | ...    | ...          | ...           | ...             | ...            | ... |
|  | 7022       | 7203-OYKCT | Male          | 0       | No         | No     | 72           | Yes           | Yes             | Fiber optic    | No  |
|  | 7023       | 1035-IPQPU | Female        | 1       | Yes        | No     | 63           | Yes           | Yes             | Fiber optic    | No  |
|  | 7034       | 0639-TSIQW | Female        | 0       | No         | No     | 67           | Yes           | Yes             | Fiber optic    | Yes |
|  | 7039       | 2234-XADUH | Female        | 0       | Yes        | Yes    | 72           | Yes           | Yes             | Fiber optic    | No  |
|  | 7042       | 3186-AJIEK | Male          | 0       | No         | No     | 66           | Yes           | No              | Fiber optic    | Yes |

1124 rows × 21 columns

## 10. Display customers who are female, senior citizens, and have churned.

```
In [74]: df.columns
```

```
Out[74]: Index(['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents',
        'tenure', 'PhoneService', 'MultipleLines', 'InternetService',
        'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport',
        'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling',
        'PaymentMethod', 'MonthlyCharges', 'TotalCharges', 'Churn'],
        dtype='object')
```

```
In [77]: df[(df['gender'] == 'Female') & (df['SeniorCitizen'] == 1) & (df['Churn'] == 'Yes')]
```

```
Out[77]:
```

|      | customerID     | gender | SeniorCitizen | Partner | Dependents | tenure | PhoneService | MultipleLines | InternetService | OnlineSecurity |
|------|----------------|--------|---------------|---------|------------|--------|--------------|---------------|-----------------|----------------|
| 53   | 7495-<br>OOKFY | Female | 1             | Yes     | No         | 8      | Yes          | Yes           | Fiber optic     | No             |
| 99   | 4598-<br>XLKNJ | Female | 1             | Yes     | No         | 25     | Yes          | No            | Fiber optic     | No             |
| 113  | 4846-<br>WHAFZ | Female | 1             | Yes     | No         | 37     | Yes          | Yes           | Fiber optic     | No             |
| 126  | 4412-<br>YLTKF | Female | 1             | No      | No         | 27     | Yes          | Yes           | Fiber optic     | No             |
| 139  | 0390-<br>DCFDQ | Female | 1             | Yes     | No         | 1      | Yes          | No            | Fiber optic     | No             |
| ...  | ...            | ...    | ...           | ...     | ...        | ...    | ...          | ...           | ...             | ...            |
| 6971 | 7240-<br>FQLHE | Female | 1             | Yes     | No         | 18     | Yes          | Yes           | Fiber optic     | No             |
| 6972 | 6664-<br>FPDAC | Female | 1             | No      | No         | 56     | Yes          | Yes           | Fiber optic     | No             |
| 6993 | 6583-<br>QGCSI | Female | 1             | Yes     | No         | 50     | Yes          | Yes           | Fiber optic     | No             |
| 7008 | 1980-<br>KXVPM | Female | 1             | No      | No         | 3      | Yes          | Yes           | Fiber optic     | No             |
| 7010 | 0723-<br>DRCLG | Female | 1             | Yes     | No         | 1      | Yes          | Yes           | Fiber optic     | No             |

240 rows × 21 columns

## 11. Group customers by InternetService and get the average MonthlyCharges.

```
In [80]: df.columns
```

```
Out[80]: Index(['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents',
        'tenure', 'PhoneService', 'MultipleLines', 'InternetService',
        'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport',
        'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling',
        'PaymentMethod', 'MonthlyCharges', 'TotalCharges', 'Churn'],
        dtype='object')
```

```
In [82]: df.groupby('InternetService')['MonthlyCharges'].mean()
```

```
Out[82]: InternetService
DSL      58.102169
Fiber optic  91.500129
No       21.079194
Name: MonthlyCharges, dtype: float64
```

## 12. Count how many customers are in each Contract type.

```
In [85]: df['Contract'].value_counts()
```

```
Out[85]: Contract
Month-to-month    3875
Two year          1695
One year          1473
Name: count, dtype: int64
```

## 13. For each PaymentMethod, calculate total and average MonthlyCharges.

```
In [88]: df.columns
```

```
Out[88]: Index(['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents',
            'tenure', 'PhoneService', 'MultipleLines', 'InternetService',
            'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport',
            'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling',
            'PaymentMethod', 'MonthlyCharges', 'TotalCharges', 'Churn'],
            dtype='object')
```

```
In [91]: df.groupby('PaymentMethod')['MonthlyCharges'].agg(['sum', 'mean'])
```

```
Out[91]:
```

|                                  | sum       | mean      |
|----------------------------------|-----------|-----------|
| <b>PaymentMethod</b>             |           |           |
| <b>Bank transfer (automatic)</b> | 103745.45 | 67.192649 |
| <b>Credit card (automatic)</b>   | 101231.85 | 66.512385 |
| <b>Electronic check</b>          | 180345.00 | 76.255814 |
| <b>Mailed check</b>              | 70794.30  | 43.917060 |

## 14. Convert TotalCharges to numeric (it may be stored as object).

```
In [97]: df.dtypes
```

```
Out[97]: customerID      object
gender                object
SeniorCitizen         int64
Partner               object
Dependents            object
tenure                int64
PhoneService          object
MultipleLines         object
InternetService       object
OnlineSecurity        object
OnlineBackup          object
DeviceProtection      object
TechSupport           object
StreamingTV           object
StreamingMovies       object
Contract              object
PaperlessBilling      object
PaymentMethod         object
MonthlyCharges        float64
TotalCharges          object
Churn                 object
dtype: object
```

```
In [101]: df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
df.dtypes
```

```
Out[101]: customerID      object
gender                object
SeniorCitizen         int64
Partner               object
Dependents            object
tenure                int64
PhoneService          object
MultipleLines         object
InternetService       object
OnlineSecurity        object
OnlineBackup          object
DeviceProtection      object
TechSupport           object
StreamingTV           object
StreamingMovies       object
Contract              object
PaperlessBilling      object
PaymentMethod         object
MonthlyCharges        float64
TotalCharges          float64
Churn                 object
dtype: object
```

```
In [103]: df.head(1)
```

Out[103..

|   | customerID | gender | SeniorCitizen | Partner | Dependents | tenure | PhoneService | MultipleLines    | InternetService | OnlineSecurity | .. |
|---|------------|--------|---------------|---------|------------|--------|--------------|------------------|-----------------|----------------|----|
| 0 | 7590-VHVEG | Female | 0             | Yes     | No         | 1      | No           | No phone service | DSL             | No             | .. |

1 rows × 21 columns

15. Remove rows with missing or blank values in TotalCharges.

In [106.. df\_clean = df.dropna(subset = ['TotalCharges'])

In [108.. df\_clean.shape

Out[108.. (7032, 21)

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